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MILITARY SPECIFICATION

TEST SET, TELEPHONE, TS-3647/G

This specification is approved for use by the Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the design, fabrication and testing necessary for the manufacture of Test Set, Telephone, TS-3647/G. This test set is also known as the Cable Orderwire Unit (COU) and is one of the electronic units comprising the Digital Group Multiplexer (DGM) family of cable system equipments developed by the Joint Tactical Command, Control and Communications Agency (JTC^3A, formerly TRI-TAC). See 6.1 for intended use of the TS-3647/G.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified the issue of these documents shall be those listed in the issue of the Department of Defense Index of Specification and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, US Army Communications-Electronics Command, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC No. N/A FSC 6625

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SPECIFICATIONS

MILITARY

MIL-P-116	Preservation-packaging, Methods of
MIL-Q-9858	Quality Program Requirements
MIL-P-11268	Parts, Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-I-45208	Inspection System Requirements
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Classification of Visual and Mechanical Defects for Equipment, Electronic, Wired and Other Devices
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461 Notice 4	Electromagnetic Interference Characteristics, Requirement for Equipment
MIL-STD-462 Notice 3	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-781C	Reliability Tests, Exponential Distribution
MIL-STD-810D	Environmental Test Methods and Engineering Guidelines
MIL-STD-2073-2	Packaging Requirements Code
MIL-STD-45662	Calibration System Requirements

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2.1.2 Other Government documents. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issue shall be those in effect on the date of solicitation.

SPECIFICATIONS

Communications-Electronics Command

CR-CX-0173-001	Nuclear Survivability/Vulnerability Provisions for the Family of Digital Group Multiplexer Equipments

DRAWINGS

Communications-Electronics Command

DL-SM-D-875937	Test Set, Telephone TS-3647/G
SM-D-884683	Circuit Card Assembly, Orderwire

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.1.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein(except for associated detail specifications, specification sheets, or MS standards) the text of this specification shall take precedence. Nothing in this specification, however, shall supersed applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article units. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.3.

3.2 Construction.

3.2.1 Fabrication. The equipment shall be constructed in accordance with DL-SM-B-875937 and the requirements specified herein.

3.2.2 Printed wiring. The printed wiring shall meet the requirements of MIL-STD-454 Requirement 17.

3.2.3 Semiconductor devices and integrated circuits. The semiconductor devices and integrated circuits shall meet the requirements of MIL-STD-454 Requirement 30 and 64.

3.3 Parts, materials, and processes general. In addition to the requirements of this specification, the requirements of MIL-P-11268 including the selection requirements therein shall apply.

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and drawing DL-SM-D-875937.

3.5 Marking. (See 4.15) The marking shall conform to MIL-M-13231, with approvals in accordance with requirements specified in the contract or purchase order.

3.5.1 Serial numbers. (See 4.16) A unique serial number is required on each nomenclatured item and CCA. The numbers shall be consecutively assigned.

3.6 Electrical preconditioning (burn-in). (See 4.6) Prior to submission to the specified quality conformance inspections, each assembled TS-3647()/G shall be electrically preconditioned for a

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minimum of 48 hours operating time. The last 24 hours of electrical preconditioning shall be failure free. Each unit fabricated shall be subjected to this process. Failure free is defined as the successful completion of the operational test of 4.7.4 during and after burn-in.

3.7 Electrical. (See 4.7)

3.7.1 Cable-side orderwire. (See 4.7.1)

3.7.1.1 Minimum output. (See 4.7.1.1) With each of the four directions terminated in 430 ohms, and a test tone of 1 kHz, the distortion at each output shall be less than 5% at an output level of +21 dBm with 9V supplies and +17 dBm with 6 volts supplies.

3.7.1.2 Gain, microphone to cable. (See 4.7.1.2) With 1.4 mV rms at 1 kHz applied to the microphone input terminals and the terminations as in 3.7.1.1 above, the TS-3647 shall output +21.0dBm ±1 dB in each direction with 9V supplies.

3.7.1.3 Gain, cable to earphone. (See 4.7.1.3) With an input on one of the four directions in the range of -24 to +26 dBm, the TS-3647 shall output from -9 to +2.5 dBm into 1000 ohms at the earphone terminals.

3.7.1.4 Ringing output. (See 4.7.1.4) With a 9V battery and terminated as in 3.7.1.1, the ringing output level shall be at least +21 dBm in each direction. The frequency shall be 1600 ±32 Hz.

3.7.1.5 Ringing sensitivity. (See 4.7.1.5) With an input on one of the four directions of -25 dBm at a frequency of 1600 ± 32Hz, the buzzer shall sound.

3.7.2 Equipment-side orderwire. (See 4.7.2)

3.7.2.1 Minimum output. (See 4.7.2.1) With power supply voltage of plus and minus 9V and a test tone level of 1 kHz, the distortion with an output of +21.3 dBm shall be less than 5%.

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3.7.2.2 Gain, microphone to 600 ohm load. (See 4.7.2.2) With 9V supplies and with 1.4 mVrms applied to the microphone input terminals at 1 kHz, the TS-3647 shall output 21.3 dBm ±1 dB into 600 ohms.

3.7.2.3 Gain, 600 ohm input to earphone. (See 4.7.2.3) When an input into the equipment-side interface in the range of -34.4 to -7.9 dBm is applied to the 600 ohm input terminals, the TS-3647 shall output from -9 to +2.5 dBm into 1000 ohms at the earphone terminals.

3.7.3 Monitoring facility. (See 4.7.3) When connected to the Pulse Form Restorer TD-1218/G, or Pulse Form Restorer TD-1219/G, the TS-3647 shall be capable of monitoring the following, using 4 green lights.

3.7.3.1 Cable current measurements. (See 4.7.3.2) The unit shall be capable of monitoring cable currents of 40 and 90 mA.

3.7.3.2 Diphase signal levels. (See 4.7.3.3) The unit shall be capable of checking for proper diphase signal level in both directions on the cable with the switch in the SIGNAL position. The diphase signal shall be detected in the Pulse Restorer and presented to the TS-3647 as a DC level. A green light indicates normal outgoing diphase signal detected.

3.7.3.3 Battery voltage check. (See 4.7.3.4) The unit shall provide means for checking the battery supply voltage.

3.7.4 Operational check. (See 4.7.4)

3.7.4.1 Maintenance OW check. There shall be no group errors for a measurement period of 10 seconds while ringing and talking over the maintenance OW from the MD-1023 to the TS-3647. The maintenance OW shall have a clear voice and ringing signal.

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3.7.4.2 Monitor facility and battery check.

Parameter	Requirement
Battery Check	All lights light
A-B Current Check	Green light
A-B Signal Check	Green light
B-A Current Check	Green light
B-A Signal Check	Green light

3.7.5 Environmental operational tests. (See 4.7.5) Satisfactory operation of environmental test units shall be verified by 3.7.4, 3.7.4.1 and 3.7.4.2.

3.8 Service conditions. (See 4.8)

3.8.1 Temperature. (See 4.8.1)

3.8.1.1 High temperature. (See 4.8.1.1) The equipment shall be operable without degradation in specified performance at ambient air temperatures as high as +48.9°C (+120°F) plus the heating effects of solar radiation at a rate of 355 BTU/ft^2/hr. The equipment shall withstand exposure to ambient air temperatures as high as +71.1°C (+160°F).

3.8.1.2 Low temperature. (See 4.8.1.2) The equipment shall be operable without degradation in specified performance at ambient temperatures as low as -45°C (-50°F). The equipment shall withstand exposure (non-operating) to ambient air temperatures as low as -57°C (-70°F).

3.8.2 Humidity. (See 4.8.2) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during and after prolonged exposure to extreme high humidity levels as encountered in tropical areas.

3.8.3 Low pressure (altitude). (See 4.8.3) The equipment shall be operable without degradation in specified performance at altitudes up to 15,000 feet above sea level, and shall withstand air transportation at altitudes up to 40,000 feet above sea level.

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3.8.4 Dust. (See 4.8.4) The equipment shall with-stand, in both operating and non-operating conditions, exposure to fine dust particles with wind speeds of 1750 ft/min (17.3 knots) and exposure to sand particles with wind speeds of 5465 ft/min (56.4 knots).

3.8.5 Salt fog. (See 4.8.5) The equipment shall be resistant to the corrosive effects of salt fog atmosphere.

3.8.6 Fungus. (See 4.8.6) The equipment, in the assembled-and-ready-for-delivery condition, shall provide no nutrients in material, coating or contaminant form or support fungal growth when tested in accordance with 4.8.6. In addition, the equipment shall operate satisfactorily after exposure to the test conditions of 4.8.6. Only inherently fungus resistant grades of materials per requirement 4, MIL-STD-454, shall be used.

3.8.7 Immersion. (See 4.8.7) Prior to and following vibration and shock testing, the equipment in its transit case shall be immersion-proof to a covering depth of 3 feet of water, for a period of not less than 2 hours.

3.8.8 Vibration and shock. (See 4.8.8)The equipment shall withstand vibration and shock induced during field transport by military vehicle and by rough handling. The equipment shall withstand vibrations encountered during transportation and shocks encountered during servicing.

3.8.9 Rain. (See 4.8.9) The equipment shall be operable without degradation in specified performance, and shall sustain no physical damage during periods of heavy precipitation with intermittent wind.

3.9 Electromagnetic interference (EMI) (See 4.9)

3.9.1 Emission. The unit, less connecting cables, shall meet the following emission requirements of MIL-STD-461A, Notice 4:

RE02	(14 kHz to 1 GHz)
RE02.1	(14 kHz to 1 GHz)

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3.9.2 Susceptibility. The unit, with connecting cables, shall meet the following susceptibility requirements of MIL-STD-461A, Notice 4:

RS02

RS03 (10 kHz to 1 GHz)

3.9.3 Bonds and grounds. (See 4.9) The unit is portable equipment and shall be grounded to the system through its interconnecting cables.

3.10 Reliability. (See 4.10) The unit shall have a specified mean-time-between-failure (MTBF), 0_O, of 10,000 hours when in the normal modes of operation.

3.11 Interchangeability. (See 4.11) Like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454.

3.12 Unit subassembly electrical performance. (See 4.12) The unit subassemblies listed below shall meet the requirements specified on their respective test requirement specification (TRS) drawings, as referenced in the following subassembly drawings:

CCA	Drawing
Orderwire	SM-D-884683

No further electrical subassembly tests are required.

3.13 Operational test. (See 4.7.4) Each fabricated and tested unit shall be subjected to a final operational test in accordance with 4.7.4 after all other inspections and tests are completed to determine that the unit is operating prior to packaging. The operational test of Group A satisfies this requirement for those units for which that is the last inspection or test.

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3.14 System safety engineering. (See 4.13)

3.14.1 Personnel hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirements 1 and 8 shall be applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly processes, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual examination and inspection.

3.14.2 Rounding exposed edges. Exposed edges and corners shall be rounded sufficiently to preclude puncture or laceration hazards.

3.14.3 Radioactive materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes/spark gaps, lenses).

3.15 Workmanship. (See 4.13) The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and with the applicable portion of MIL-P-11268 as follows:

a. Adhesives

b. Cables, cords and wires, electrical

c. Wire (harnessing)

d. Plastic material and parts

e. Rubber

f. Wiring and cabling

g. Connectors (electrical)

h. Printed wiring assemblies

i. Processes

3.16 Nuclear survivability. The unit shall be designed to withstand the Initial Nuclear Radiation requirements of Section 3 of CR-CX-Q173-001..

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed measurements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the cotractor's overall inspection system or quality program. The absence of any inspecton requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection. (See 4.3) Does not include packaging.

b. Inspections covered by subsidiary documents. (See 4.4).

c. Quality conformance of inspection before packaging (see 4.5).

d. Packaging inspection (see 4.14).

4.3 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.3.1 First article units. The contractor shall furnish a minimum of 18 first article units of the complete Test Set, Telephone TS-3647/G.

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4.3.2 First article inspection. The first article inspection shall consist of inspections specified in table I and shall be performed in the order specified therein except that all hardwired and plug-in assemblies shall undergo in process Group A inspection prior to burn-in.

Notes to table I:

a. The inspections 1 through 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by Inspections 4 and 5 specify the order of inspections for the indicated unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after this inspection before proceeding with subsequent inspection.

d. The quantity of units shall be as specified in the contract or purchase order. All first article units shall be subjected to Group A inspection.

4.3.3 First article inspection data. The first article test plan and test report(s) shall be as required in the contract or purchase order.

4.4 Inspections covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

	Where required
Printed Wiring	3.2.2
Semiconductor Devices	3.2.3
Parts, materials, and processes	3.3
Finish	3.4
Marking	3.5

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TABLE I. First article inspection.

Inspection	Rqmt Para	Test Para	Unit 1	2	3	4	5	6	7	8-19	20+
1. Inspections by Subsidiary Documents	NA	4.4	Inspections to be performed on all units.								
2. Burn-In	3.6	4.6	Inspections to be performed on all units.								
3. Group A Inspection	(See Table II)		Inspections to be performed on all units.								
4. Group C Inspection											
Subgroup 1											
High Temperature	3.8.1.1	4.8.1.1	1								
Low Temperature	3.8.1.2	4.8.1.2	2								
Subgroup 2											
EMI	3.9	4.9		1							
Immersion	3.8.9	4.8.9			1						
Vibration	3.8.7.1	3.8.7.1			2						
Shock	3.8.7.2	3.8.7.2			3						
Immersion	3.8.9	4.8.9			4						
Subgroup 3											
Humidity (Note c)	3.8.2	4.8.2				1					
Rain (Note c)	3.8.8	4.8.8				2					
Subgroup 4											
Altitude	3.8.3	4.8.3					1				
Dust (Note c)	3.8.4	4.8.4					2				
Salt Fog (Note c)	3.8.5	4.8.5					3				
Fungus (Note c)	3.8.6	4.8.6						1			
Subgroup 5											
Dimensional Interch	3.11	4.11							1		
5. Group D Inspection											
Reliability	3.10	4.10								12 Units	

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4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.4. and 4.5.1 through 4.5.5. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

4.5.1 Group A inspection. Each unit on contract or purchase order and each subassembly and assembly identified in Table II shall be inspected for conformance to the inspections specified in table II.

4.5.1.1 Order of inspection within Group A. Group A inspection shall be performed in the order listed in table II. The order of inspection of the subassemblies, assemblies, and units (and equipment) shall be as follows:

a. Prior to their assembly into the end equipment, the assemblies and subassemblies identified in table II shall be subjected to a 100% visual/mechanical and electrical inspection by the contractor for conformance with the inspection requirements provided in table II.

b. Only for spares that are not installed into complete units, do the following: After completion of the above 100% inspection by the contractor, discrete lots shall be formed from the subassemblies and assemblies which have passed the 100% inspection and these shall be subjected to sampling inspection utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in table II (Quality Assurance Verification). Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105.

c. Assemblies from the discrete lots which have passed the Quality Assurance Verification Inspection of step b. above shall be assembled into the end equipments, which shall then be subjected by the contractor to the burn-in test requirements of 4.6 and a 100% visual/mechanical and electrical inspection in accordance with table II in that order.

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d. After completion of the above 100% inspection by the contractor, discrete lots shall be from units which passed the 100% inspection and these shall be subjected to sampling inspection by the contractor utilizing the procedures of MIL-STD-105 using the general inspection levels and AQLs indicated in table II. Group A electrical inspection shall be performed in the order shown in table II.

TABLE II. Group A inspection.

Inspection	Reqt Para	Insp Para	AQL Major	Minor
VISUAL-MECHANICAL (Including Spare Assemblies)				
CCA Orderwire	3.15	4.12	2.5%	4%
Case and Covers	3.15	4.12	2.5%	10%
ELECTRICAL PERFORMANCE (Including Spare Assemblies)				
CCA Orderwire	3.12	4.13	1.0%	*
ELECTRICAL				
Cable-Side Order-wire Tests	3.7.1	4.7.1	1.0%	*
Equipment-Side Orderwire Tests	3.7.2	4.7.2	1.0%	*
Monitoring facility	3.7.3	4.7.3	1.0%	*
Operational Checks	3.7.4	4.7.4	1.0%	*

*All Electrical defects are Major.

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4.5.2 Group C inspection. Group C inspection shall be performed on units that have passed Group A inspection. The inspection shall consist of the inspections specified in table III. Samples shall be selected in accordance with 4.5.3.1.

4.5.2.1 Sampling for Group C inspection.

4.5.2.1.1 Subgroup 1. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every third month. The unit shall be subjected to high and low temperature in that order.

4.5.2.1.2 Subgroup 2. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fourth month, and one unit shall be selected every eighth month. The unit selected every fourth month shall be subjected to immersion, vibration, shock and immersion testing with the unit subjected to testing in that order. The unit selected every eighth month shall be subjected to EMI/EMC testing.

4.5.2.1.3 Subgroup 3. For this subgroup, one unit shall be selected from the first production lot. For subsequent Group C inspection, one unit shall be selected every fifth month. The unit shall be subjected to humidity and rain testing.

4.5.2.1.4 Subgroup 4. For this subgroup, two units shall be selected from the first production lot. For subsequent Group C inspection, two units shall be selected every twelfth month. One unit shall be subjected to Altitude, Dust and Salt Fog testing and one unit shall be subjected to Fungus testing.

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TABLE III. Group C inspections.

Inspection	Requirement Paragraph	Inspection Paragraph
Subgroup 1		
High Temperate	3.8.1	4.8.1.1
Low Temperate	3.8.1	4.8.1.2
Altitude	3.8.3	4.8.3
Subgroup 2		
Immersion	3.8.9	4.8.9
Vibration	3.8.7.1	4.8.7.1
Shock	3.8.7.2	4.8.7.2
Bounce, Loose Cargo		4.8.7.2.1
Drop		4.8.7.2.2
Immersion	3.8.9	4.8.9
EMI	3.9	4.9
Subgroup 3		
Humidity	3.8.2	4.8.2 (Note 1)
Rain	3.8.8	4.8.8
Subgroup 4		
Altitude	3.8.3	4.8.3
Dust	3.8.4	4.8.4 (Note 1)
Salt Fog	3.8.5	4.8.5 (Note 1)
Fungus	3.8.6	4.8.6 (Note 1)
Subgroup 5		
Dimensional Interch	3.11	4.11

NOTE 1: The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent test.

4.5.2.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in 4.5.3.1.

4.5.2.3 Group C failure. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

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4.5.3 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass Group A inspection only.

4.5.4 Group D inspection. This inspection shall consist of the tests specified in table IV and shall be performed on units from lots which have been subjected to and have met Group A inspection. Test conditions, methods, and procedures shall be in accordance with 4.10.

TABLE IV. Group D inspection.

Inspection	Requirement Paragraph	Inspection Paragraph
Reliability	3.10	4.10

4.5.4.1 Sampling for Group D inspection. Twelve units shall be selected at random from the first month's production and subjected to the test of 4.10. For subsequent Group D inspection, twelve units shall be selected at random from every twelfth month's production.

4.5.4.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order.

4.5.4.3 Reinspection of conforming Group D sampling units. Unless otherwise specified, sample units which have been subjected to and have passed Group D inspection may be accepted on the contract provided all damage is reworked to specification requirements and the sample units are resubjected to and pass Group A inspection only.

4.6 Electrical preconditioning (burn-in). (See 3.6) During electrical preconditioning, the equipment shall be connected per figure 1. Test conditions of 4.10.3 shall apply except that the duty cycle shall be 100 percent. If a failure occurs during this preconditioning, the failure shall be recorded, the failed component(s) replaced, the preconditioning cycle reinstituted. The requirements of 3.6 shall be met.

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4.6.1 Burn-in test. Connect the equipment as shown in Figure 1. Turn on power and run the unit for 48 hours. Set the METER switch to INCOMING SIG. Monitor the cable current using the TS-3647 SELECT switch on the front panel; verify that it is in the "green zone". At the beginning and end of the burn-in, and at the beginning and end of each normal workday, press the RING button on the MD-1023 and verify that the buzzer in the case sounds. Then press the RING button on the case and verify that the CALL lamp on the MD-1023 lights.

4.7 Electrical. (See 3.7) Unless otherwise specified all tests shall be performed at ±9 Vdc. Tests shall be conducted at room ambient temperature, pressure, and humidity per MIL-STD-810 unless otherwise specified. The electrical test equipment listed in tables V and VI will be utilized during testing and will be calibrated in accordance with MIL-STD-45662A. The standard test equipment shall be provided by the contractor. Government technical representative approval is required prior to use of proposed substitute items. The Government will be responsible for supplying the Government Furnished Equipment (GFE) listed in table VII. Table VIII identifies the external interface connections used in the test figures. Unless otherwise noted all resistors in the test figures shall be .5 watts.

TABLE V. Standard test equipment.

Description	Mfg./Model No. (or equivalent)
Oscilloscope	Tektronix 454
DC Power Source	H.P. 6205B
Audio Oscillator	H.P. 200CD
AC Voltmeter	H.P. 400L
Attenuator, Audio	Shallco Model 790

TABLE VI. Special test equipment.

Equipment Name	Drawing No.	Function Description
Group Simulator	A3051667	The Group Simulator simulates the group data, orderwire and telemetry signals. It generates either a long or short pseudo-random sequence. The three signals can be generated simultaneously or in any combination. A TTL error output signal is provided for counting errors externally.

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TABLE VII. Government furnished equipment (GFE).

Equipment	Description
CX-11230()/G Cable	Twin coax supplied as GFE for test (1/4 mile reels)
MD-1023(P)/G	Low Speed Cable Driver Modem
TD-1218/G	Low Speed Pulse Restorer
TS-3647/G	Test Sets, Telephone
H-250 Headset	For Voice Communication between TS-3647/G and MD-1023 (P)/G

Table VIII. Control orderwire unit interface.

Function	Pin
AVOW to TD-1234 or TD-1233	P, R
AVOW from TD-1234 or TD-1233	K, L
A-B Current Monitor High	A
Return	B
A-B Output Monitor	G
B-A Current Monitor High	D
Return	C
B-A Output Monitor	H
GND	E, F
Audio XMT	D
Audio RCV	B
PTT	C
GND	A

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4.7.1 Cable-side orderwire. (See 3.7.1)

4.7.1.1 Minimum output. (See 3.7.1.1)

a. Connect the equipment as shown in figure 2 (A).

b. Adjust the power supply to 9.0 Vdc. Turn UUT power switch ON.

c. Close the PTT switch. Using the VTVM, monitor the output voltage across the two 215-ohm terminations at the OW connector. Adjust the output of oscillator at 1 kHz to obtain a voltage of 7.35 Vrms at the OW connector (this is equivalent to putting +21 dBm into 430 ohms).

d. Disconnect the VTVM from the OW connector and connect the distortion analyzer. Measure the distortion. Verify that the signal has less than 5 percent distortion.

e. Repeat steps a thru d with the unit power set to 6.0 volts DC and the oscillator set for 4.65 Vrms at the OW connector. (This is equivalent to +17 dBm into 430 ohms.)

4.7.1.2 Gain: microphone to cable. (See 3.7.1.2)

a. Connect the equipment as shown in figure 2(A).

b. Adjust the audio oscillator output voltage such that the input voltage to the UUT is 1.4 mV at 1 kHz.

c. Close the PTT switch. Measure the output voltage across the two 215-ohm terminations at the orderwire connector.

d. Verify that the outputs are +21 dBm $\pm$1 dB into 430 ohms (6.56 to 8.26 volts).

4.7.1.3 Gain: cable to earphone. (See 3.7.1.3)

a. Connect the equipment as shown in figure 2 (B) with the oscillator connected to pin A of the orderwire connector.

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b. Adjust the oscillator at 1 kHz for an output of 0.041 Vrms, equivalent to -24 dBm into 430 ohms at the OW connector.

c. Measure the voltage across the 1000-ohm load at the Handset connector.

d. Adjust the oscillator at 1 kHz for an output of 13.1 Vrms, equivalent to +26 dBm, into 430-ohms at the orderwire connector.

e. Measure the voltage across the 1000-ohm load at the Handset connector.

f. Verify that the requirements of 3.7.1.3 are met.

g. Repeat steps b through f with the oscillator connected to pin D of the orderwire connector.

4.7.1.4 Ringing output. (See 3.7.1.4)

a. Connect the equipment as shown in figure 2 (A), disconnect the oscillator.

b. Set the power supply voltage to 9.0 Vdc and open the PTT switch.

c. Press and hold the RING button on the UUT.

d. Measure the signaling voltage across the two 215-ohm load at the OW connector.

e. Disconnect the VTVM and connect the counter. Measure the signaling frequency.

f. Verify that the requirements of 3.7.1.4 are met.

4.7.1.5 Ringing sensitivity. (See 3.7.1.5)

a. Connect the equipment as shown in figure 2 (B), disconnect the VTVM.

b. Adjust the oscillator frequency to 1568 Hz and the level to 37 mV. UUT buzzer should sound.

c. Adjust the oscillator frequency to 1632 Hz and the level to 37 mV. UUT buzzer should sound.

d. Rotate the volume control, verify a change in volume.

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4.7.2 Equipment-side orderwire. (See 3.7.2)

4.7.2.1 Minimum output. (See 3.7.2.1)

a. Connect the equipment as shown in figure 3(A).

b. Adjust the power supply to 9.0 Vdc.

c. Adjust the Audio Oscillator set at 1 kHz to obtain an output voltage of 9.0 Vrms across the 600-ohm resistor at the ORDERWIRE connector on the UUT (equivalent to +21.3 dBm into 600 ohms).

d. Disconnect the voltmeter from the ORDERWIRE connector on the UUT and connect the distortion analyzer. Verify that the signal has less than 5 percent distortion.

4.7.2.2 Gain: microphone to 600-ohm load. (See 3.7.2.2)

a. Connect the equipment as shown in figure 3(A).

b. Adjust the power supply to 9.0 Vdc.

c. Adjust the audio oscillator output to 1.4 mV at 1 kHz.

d. Measure the voltage across the 600-ohm termination at the ORDERWIRE connector on the UUT.

e. Verify that the requirements of 3.7.2.2 are met.

4.7.2.3 Gain: 600-ohm input to earphone. (See 3.7.2.3)

a. Connect the equipment as shown in figure 3(B).

b. Set the power supply voltage to 9.0 Vdc.

c. Adjust the oscillator at 1 kHz for 14.8 mV (equivalent to -34.4 dBm into 600 ohms) at the ORDERWIRE connector.

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d. Measure the output across the 1000-ohm load at the Handset connector.

e. Adjust the oscillator at 1 kHz for 312 mV output (equivalent to -7.9 dBm into 600 ohms) at the ORDERWIRE connector.

f. Measure the output across the 1000-ohm load at the Handset connector.

g. Verify that the requirements are met.

4.7.3 Monitoring facility. (See 3.7.3)

4.7.3.1 Cable current measurements. (See 3.7.3.1)

a. Connect the equipment as shown in figure 4. The auxilliary TS-3647 audio output pins P and R are connected through attenuator 1 set for 26 dB attenuation. The auxilliary unit audio in-put pins K and L are connected through attenuator 2 set for 4 dB attenuation.

b. Set the power supply to 9.0 Vdc and connect to the battery terminal inside the battery compartment.

c. Turn both MD-1023 CABLE CURRENT switches to ON and verify that the UUT A to B and B to A TEST CUR indicators are illuminated.

d. Turn MD-1023 #1 CABLE CURRENT off and verify that the UUT A to B indicator extinguishes.

e. Turn MS-1023 #2 CABLE CURRENT off and verify that the UUT B to A indicator extinguishes.

4.7.3.2 Diphase signal levels. (See 3.7.3.2)

a. The test set-up is the same as 4.7.3.1.

b. Turn on the CABLE CURRENT switches for both MD-1023's. Set MD-1023 #1 CABLE TEST SELECT to FLT LCT and press the PUSH TO FLT LCT switch. Verify that the TEST A to B SIG indicator extinguishes until the MD-1023 displays a 2.

c. Set MD-1023 #2 CABLE TEST SELECT to FLT LCT and press the the PUSH TO FLT LCT switches. Verify that the TEST B to A SIG indicators extinguishes until MD-1024 displays a 2.

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4.7.3.3 Battery voltage check. (See 3.7.3.3)

a. Connect the equipment as shown in figure 4 with both MD-1024 cable current switches ON.

b. Adjust the DC power supply connected to the battery terminals to 6.5 Vdc and 9.0 Vdc.

c. At each voltage with the System/Battery switch in the battery position verify that A-B and B-A TEST CUR and SIG indicators illuminate.

d. Reduce the power supply voltage to 5.0 Vdc. Verify that the indicators do not illuminate.

4.7.4 Operational check. (See 3.7.4)

4.7.4.1 Maintenance OW check.

a. Connect the equipment as shown in figure 4 with connections and attenuator settings as described in 4.7.3.1 steps a and b. Set the Group Simulator and MD-1023 group rate to 14 (4608 kbps).

b. While pressing the auxilliary TS-3647 ring button, verify that no group errors are generated for 10 seconds.

c. While talking betweeen the UUT and auxilliary TS-3647, verify that the voice quality is understandable and no group errors are generated.

4.7.4.2 Monitor facility and battery check. The monitor facility and battery operation shall be verified using 4.7.3.1 through 4.7.3.3.

4.7.5 Environmental operational test. (See 3.7.5) The test set-up and procedures are shown in 4.7.4.

4.8. Servicee conditions tests. (See 3.8)

4.8.1 Temperature. (See 3.8.1)

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4.8.1.1 High temperature. (See 3.8.1.1) The equipment shall be subjected to the test of MIL-STD-810D, Method 501.2, Procedure II for cycling exposure. In step 12, raise the internal chamber temperature to +49°C (+120°). Maintain internal chamber temperature for six hours at +49°C (+120°F). Raise the internal chamber temperature to +71.1°C (+160°F) within a time period of one hour and maintain at that temperature for four additional hours. Lower the internal chamber temperature to +49°C (+120°) within a time period of one hour. Repeat the above steps two additional times to make a total of three 12-hour cycles. Step 14 shall be performed as follows: adjust the internal chamber temperature to +71.1°C (+160°F) and maintain this temperature until temperature stabilization of the test item is reached. Operate the test item until the item is stabilized.

4.8.1.2 Low temperature. (See 3.7.1.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 502.2, Procedure I. Lower the internal chamber temperature in step 2 to -51°C (-60°F) for a period of not less than two hours following stabilization of the equipment, or until 24 hours have elapsed, whichever occurs first. In step 5 of Procedure I, perform Procedure II - Operation (paragraph II-3.2 of Method 502.2). Chamber temperature in step 1 of Procedure II shall be adjusted to -45°C (-50°F).

4.8.2 Humidity. (See 3.8.2) The equipment shall be subjected to the test of MIL-STD-810D, Method 507.2, Procedure III. After completion of Procedure III, the test item shall be subjected to Procedure II. In step 4, cycle 5 of table 507.2-I shall be used.

4.8.3 Altitude The equipment shall be subjected to the test of MIL-STD-810D, Method 500.2, Procedure I. For step 2, adjust the chamber pressure to 141 mm of Hg (5.56 inches of Hg or 40,000 feet above sea level), at a rate not to exceed 2,500 fpm. The equipment shall also be subjected to the test of MIL-STD-810D, Method 500.2, Procedure II. The chamber pressure in step 2 shall be adjusted to 429 mm ofHg (16.9 inches of Hg or 15,000 feet above sea level) at a rate not to exceed 2,500 fpm.

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4.8.4 Dust. (See 3.8.3) The equipment shall be subjected to the test of MIL-STD-810D, Method 510.2, Procedure I. The air velocity in step 2 shall be 1750 ± 200 feet per minute. In step 5, reduce the air velocity to 300 ± 200 feet per minute and adjust the air temperature to + 63°C(+145°F). The equipment shall be reoriented periodically to equally expose all sides of the equipment to the dust stream. Equipment shall be operated before step 1, during step 8 and during step 11. There shall be no degration in performance as a result of the dust test, and there shall be no binding or blocking of moving parts, non-operation of contacts or relays, clogging or dust accumulation which could form the nucleus for the collection of water vapor as a result of dust penetration of the test item. The equipment shall also be subjected to the test of MIL-STD-810D, Method 510.2, Procedure II. The air velocity in step 2 shall be 5465 + 250 feet per minute. The equipment shall be reoriented periodically, such that each side of the equipment is exposed to the sand stream for 90 minutes. Equipment shall be operated before step 1, during step 5, and during step 7. There shall be no degradation in performance as a result of the sand test and there shall be no abrasion or clogging which affects the operation of the equipment. There shall be no sand penetration of the equipment as a result of the sand tests.

4.8.5 Salt fog. (See 3.8.5) The equipment shall be subjected to the test of MIL-STD-810D, Method 509.2, Procedure I. In step 2, the 48 hour period shall be constant wetting. Delete step 3. Operation in step 4 shall take place after gentle wash in running water not warmer than +38°C (+100°F) and drying. There shall be no corrosion of finishes or metals. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted, blistered or otherwise loosened finish on metal surfaces. There shall also be no clogging or binding of moving mechanical parts.

4.8.6 Fungus. (See 3.8.6) The equipment in the assembled-and-ready-for-delivery-condition, or representative samples subject to approval of the procuring agency, shall be subjected to the test of MIL-STD-810D, Method 508.3. After 28 days, based on visual examination, the equipment or samples shall show no more than sparse microbial growth with restricted tuberal growth development in an area 10% or less of the total area and no more than six unrelated minute colonies with mycelial development in non-critical areas. Critical areas include: terminal spacing, printed circuit boards, etc. The equipment shall fail if it does not meet operational performance criteria or if it shows more than the growth specified above.

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4.8.7 Immersion. (See 3.8.7) The equipment shall be subjected to the test of MIL-STD-810D, Method 512.2, Procedure I.

4.8.8 Vibration and shock. (See 3.8.8)

4.8.8.1 Vibration. The equipment shall be subjected to the test of MIL-STD-810D, Method 514.3 for Category 1 -Basic Transportation. The equipment in its transit case, if any, as restrained and/or tied down in its combat transport configuration shall be subjected to the random vibration environment for mission/field transportation. The vibration environment shall be simulated using the test schedules given in figures 514.3-4, 514.3-5, 514.3-6, and figures 514.3-10 through 514.3-22. Test duration for figures 514.3-4, 514.3-5, and 514.3-6 shall be 48 minutes per axis, for a test time of 144 minutes. Test duration for figures 514.3-10 through 514.3-22 shall be as given in the test schedules, for a test time of 3 hours. Total test time shall be 324 minutes. The test item shall not be operated during this test. The equipment in its transit case, if any, shall also be subjected to the test of MIL-STD-810D, Method 514.3 for transpsortation vibration, Category 3 -Loose Cargo Transport. Prior to and following the tests for both Category 1 and Category 3, the equipment shall be operated in accordance with MIIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance. The equipment, including transit case, shall not sustain any mechanical damage.

4.8.8.2 Transit drop. The equipment in its transit case, if any, shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure IV. Prior to and following this test, the equipment shall be operated in accordance with MIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance. The equipment, including transit case, shall not sustain any mechanical damage.

4.8.8.3 Bench handling. The equipment shall be subjected to the test of MIL-STD-810D, Method 516.3, Procedure VI. Prior to and following this test, the equipment shall be operated in accordance with MIL-STD-810D, General Test Performance Guidance, paragraph 4.5. The equipment shall not have any degradation in performance and shall not sustain any mechanical damage.

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4.8.9 Rain. (See 3.8.9) The equipment shall be subjected to the test of MIL-STD-810D, Method 506.2, Procedure I. In step 1, the test item shall be exposed to a simulated rain at a rate of 2 ± 0.5 inches per hour for 10 minutes. The rate of rainfall shall then be raised to 5 ± 1 inches per hour and held at this rate for 5 minutes. The rate shall then be reduced to 2±0.5 inches per hour for the next 15 minutes. Step 2 shall be changed as follows: starting 5 minutes after the initiation of the rain, the wind source shall be turned on and adjusted to produce a horizontal wind velocity of 40 miles per hour (3,500 feet per minute). The test item shall be operated in step 3.

4.9 EMI tests. (See 3.9) The unit shall be tested for compliance with 3.9. Test methods and procedures shall be in accordance with MIL-STD-462, Notice 3, and a Government approved test plan.

4.9.1 Bonds and grounds. (See 3.9.3) Prior to performance of EMI testing, compliance with requirements of 3.9.3 shall be ascertained. Bonding resistance measurements shall be recorded and included in the EMI test report.

4.9.2 General test conditions. During susceptibility testing, conducted and radiated emissions shall be measured with equipment operating to monitor both low speed and high speed cable systems or equipment data generator. Degradation of performance criteria shall include TS-3647's accurate measurement of cable current and voltage and operation of the maintenance orderwire according to performance characteristics of Section 3.2.1.

4.10 Reliability testing. (See 3.10)

4.10.1 First article reliability test. The first article reliability testing shall be in accordance with Test Plan VIC, of MIL-STD-781C.

4.10.2 Quality conformance (Group D) reliability test. The reliability test for Group D inspection shall be in accordance with Test Plan VIC of MIL-STD-781C.

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4.10.3 Reliability test conditions. Reliability testing shall be accomplished subject to the following conditions:

a. Duty cycle: Apply prime power to equipment under test for 4 hours. Turn power OFF for 2 hours. Turn power ON for 16 hours. Turn power OFF 2 hours. Repeat this sequence for each 24-hour cycle.

b. Chamber temperature shall be 50°C+5°-0°C (122°F to 131°F)

c. Vibration shall be 2.2G ±10% peak acceleration value at any nonresonant frequency between 20 and 60 CPS measured at the mounting points on the equipment. The duration of vibration shall be at least 10 minutes during each hour of equipment operating time.

4.10.4 Reliability test method. The detailed test procedures and conditions for performing the tests outlined in 4.10.1 and 4.10.2 shall be in accordance with a contractor prepared, Government approved test plan.

4.10.5 Failure definition. A failure is defined as any malfunction which causes or may cause inability to commence operation, cessation of operation, or degradation below limits specified in 3.7.4.

4.10.6 Failure reporting and analysis. A failure reporting and analysis system shall be implemented per MIL-STD-781. Final determination of the failure relevancy and multiple and pattern failure categorization shall be the responsibility of the procuring activity.

4.11 Interchangeability. (See 3.11) The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.11. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

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b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of switches, sockets and fasteners.

d. Location of panel-mounting screw holes on basic unit for mating with case.

e. Location of connectors, locking pins, fasteners slides, and mountings which receive mating parts or plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

4.12 Electrical subassembly tests. (See 3.12) Plug-in sub-assemblies, including spares, shall meet the requirements of their respective test requirements specification drawings listed in 3.12.

4.13 Visual and mechanical. (See 3.14 and 3.15) The equipment shall be examined for the defects listed in MIL-STD-252 to determine compliance with 3.15. An inspection shall be performed to verify compliance with those portions of 3.14 (Systems safety engineering) which can be determined visually.

4.13.1 System Safety Design Verification Check List. An inspection shall be performed, on one unit only, during first article testing, utilizing SEL Form 1183, to verify compliance with 3.14 (System safety engineering).

4.14 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging.

b. Quality Conformance Inspection of Packaging.

4.14.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.14.2 Quality Conformance Inspection of Packaging.

4.14.2.1 Material inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

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4.14.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.14.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in table IX, "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0.

4.15 Markings. The equipment shall be inspected to verify that the markings conform to MIL-M-13231.

4.16 Serial numbers. The equipment shall be inspected to verify that the serial numbers appear in each nomenclatured item and CCA.

4.17 Nuclear tests. Tests for nuclear survivability shall be performed on the equipment in accordance with Section 3 of CR-CX-0173-001.

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TABLE IX. Packing inspection provisions.

NO.	CHARACTERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification incorrectly applied, omitted	Visual
109	Marking omitted, incorrect or illegible	Visual

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5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, table I through VII, codes as follows:

5.1.1 Level A. 3Q-1-00-00-NR-X-ED

5.1.2 Level B. 2M-1-00-00-NR-X-ED

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The TS-3647/G is a field portable test set for low and high speed coaxial cable systems developed by JTC^{3}A It performs the following functions:

Provides analog orderwire termination for TD-1218, TD-1219, TD-1233 and TD-1234.

Monitors levels of conditioned diphase signals at repeaters.

Monitors power feed current at repeater.

Checks voltage of internal 9V battery.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification and any amendment thereto.

b. Level A or B preservation and packing (see Section 5).

c. When first article rough handling tests are not required.

d. Quantity of first article units required.

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e. Place of final inspection.

f. Technical literature required.

g. Quantity of tools and running spare parts required.

h. Group C failure requirement.

i. MIL-Q-9858 or MIL-I-45208.

j. Length of cables (if any are required).

k. When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Final inspection. (See 4.5.3) Approval to ship may be withheld at the discretion of the Government pending the decision from the contracting officer on the adequacy of corrective action.

6.4.1 Initial shipments. Initial shipments of end items will be contigent upon successful completion of all first article tests.

6.5 Location of operational inspection. (See 4.7.4) It is desirable that the operational system test be performed at a location that will minimize handling after this inspection is completed. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.6 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contrator's quality control system and the quality history of the product.

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6.7 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage, and handling.

6.8 Government-furnished property. The contracting officer should arrange to furnish the property listed in table VI.

6.9 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

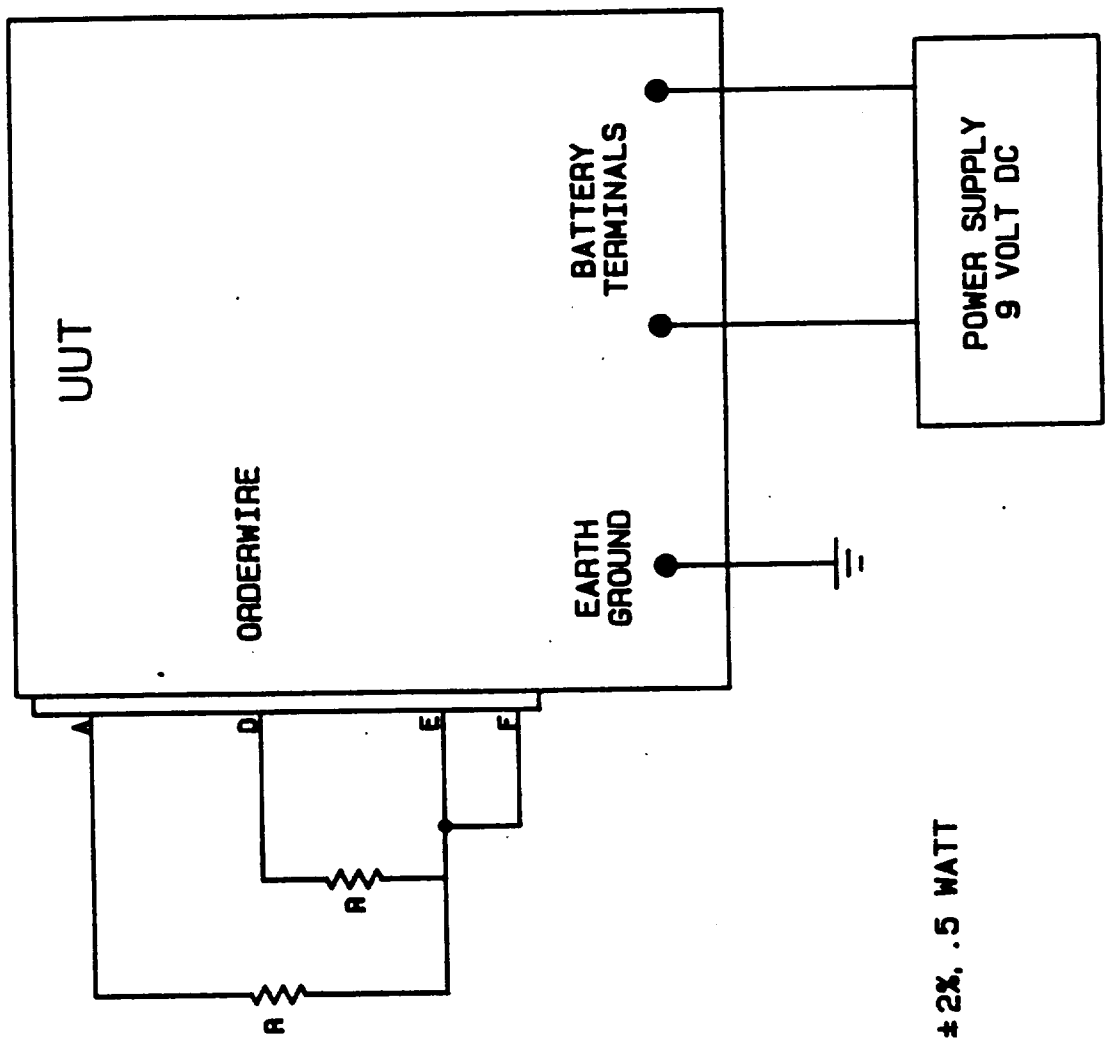
6.10 Subject terms (key-words) list.

Cable Orderwire Unit
Cable-side Orderwire
Conditioned Diphase
Digital Group Multiplex
Diphase
Maintenance Orderwire
Monitoring
Pulse Restorer
Repeater
Test Set
Test Set, Telephone
Test Set, Telephone, TS-3647/G

CUSTODIAN:
ARMY-CR

Preparing Activity
ARMY-CR
Project Nr. 5820-A834

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R 215 OHMS ±2%, .5 WATT

FIGURE 1. Burn-in.

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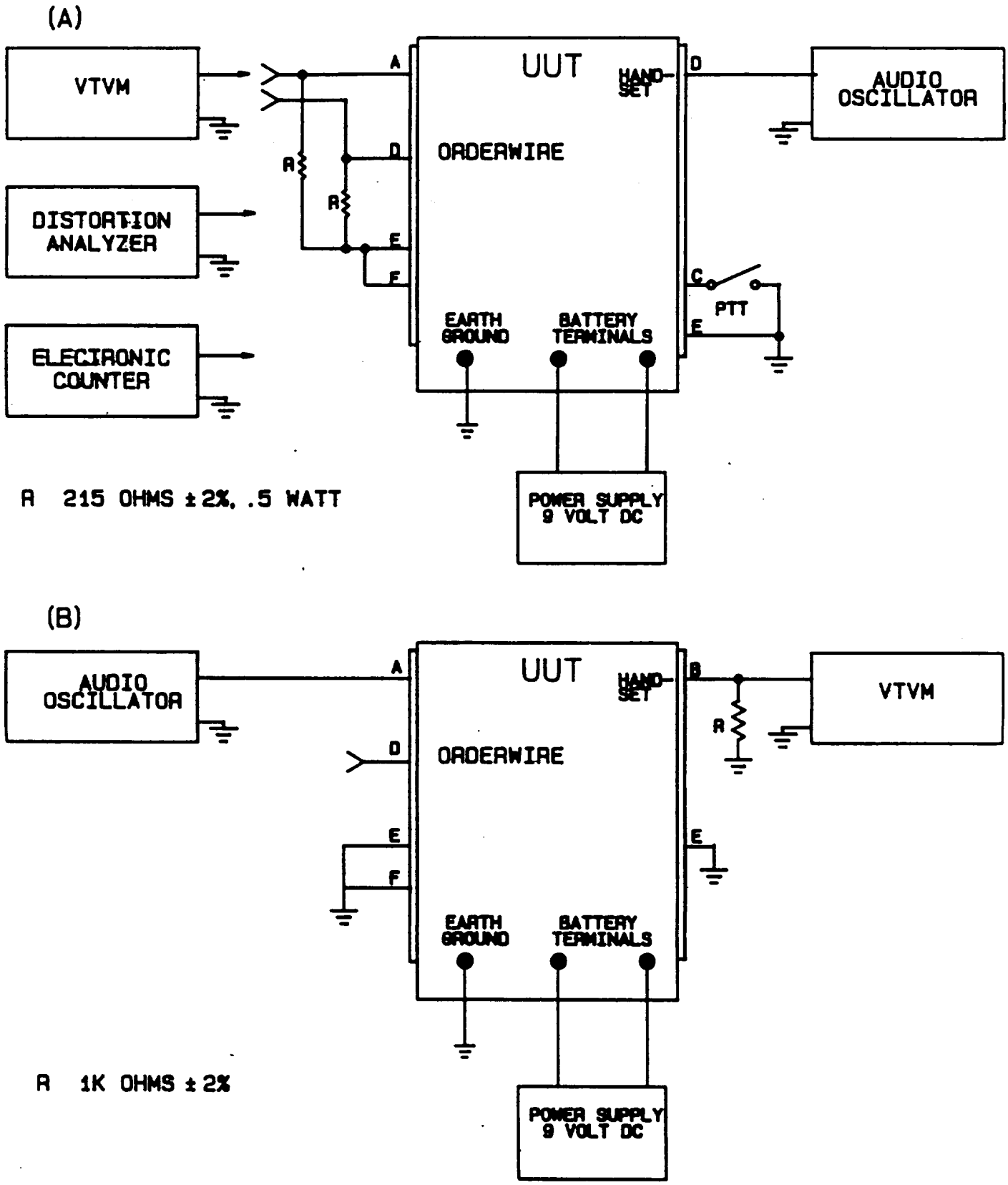


FIGURE 2. Cable-side orderwire.

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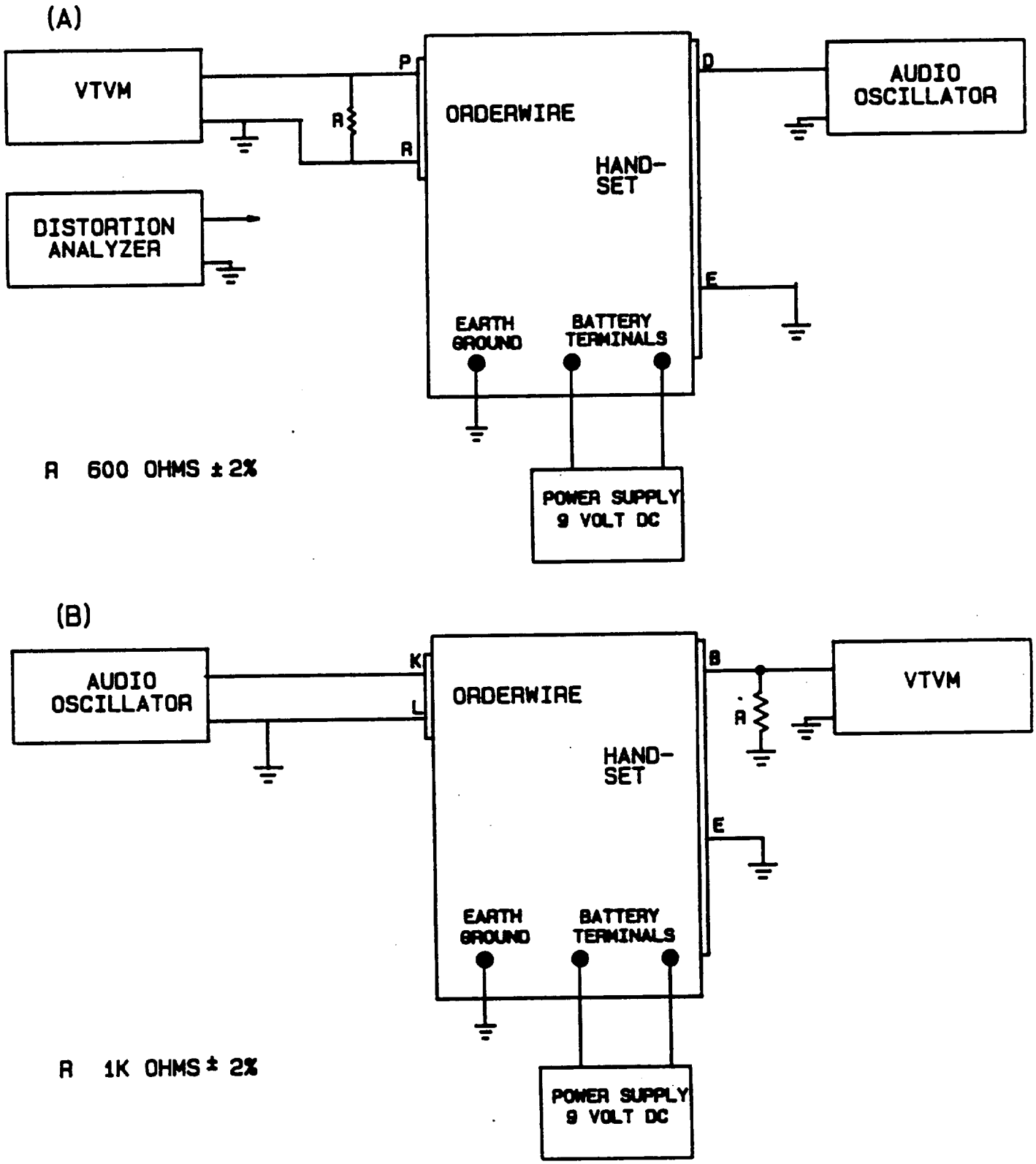


FIGURE 3. Equipment-side orderwire.

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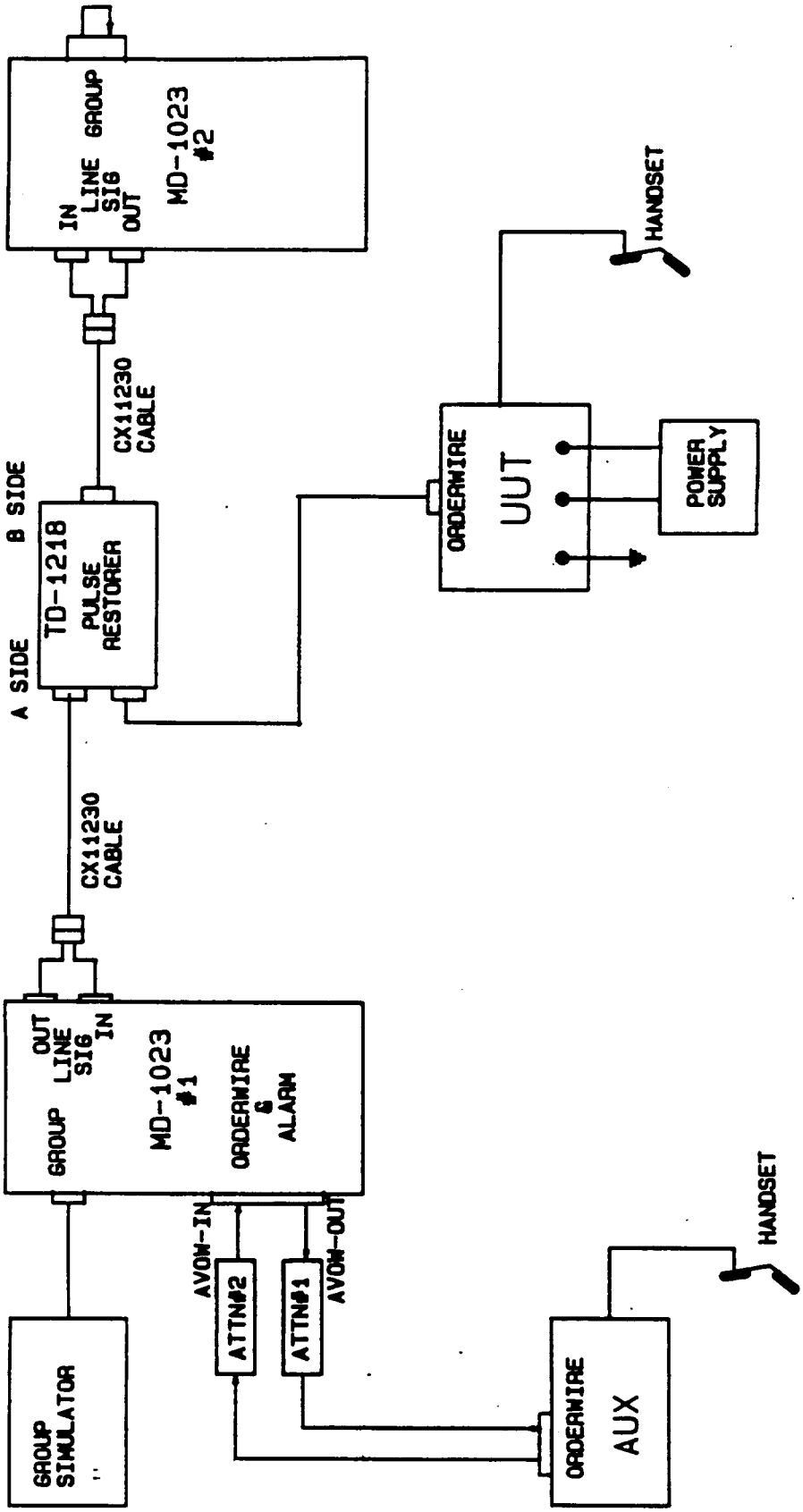


FIGURE 4. Monitor facility.

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